King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 251 (Version B)

Original exam date: 12 November 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If (a, b, c) is the point of intersection between the lines

$$L_1 : \ x = 1 + 2t, \ y = 1 - 3t, \ z = 2 - 5t$$

$$L_2 : \ x = 9 - 3s, \ y = -10 + 4s, \ z = -17 + 7s$$

then $abc =$

(a) -15

(b) 12

(c) 9

(d) 18

(e) -16

2. If $f(x, y, z) = x^3 - 4xy + 3y^3 + xy^2z^2$, then

$$f_{xyzz}(2, 2, 2) + f_{yzxz}(2, 2, 2) =$$

(a) 10

(b) 14

(c) 16

(d) 2

(e) 8

3. An equation for the level surface of $f(x, y, z) = 4x^2 + 4y^2 - z^2$ that passes through the point $(0, 0, 3)$ is given by

(a) $z^2 = x^2 + y^2 + 9$

(b) $z^2 = 4x^2 + 4y^2$

(c) $z^2 = x^2 + y^2 + 3$

(d) $z^2 = 4x^2 + 4y^2 + 9$

(e) $z^2 = 4x^2 + 4y^2 + 81$

4. Which one of the following statements is FALSE about the plane

$$3x - 2y + z = 6?$$

(a) It is parallel to the plane $6x - 4y - 2z = 10$

(b) It intersects the x-axis at $(2, 0, 0)$

(c) It passes through the point $(1, 1, 5)$

(d) The point $(0, -3, 0)$ lies in the plane

(e) It has a normal vector $\vec{n} = \langle 3, -2, 1 \rangle$

5. If $w = 2x^2 - y^2$, $x = s\cos t$, $y = s\sin t$, then $\dfrac{\partial w}{\partial t} =$

(a) $-3s^2\cos(2t)$

(b) $-3s^2\sin(2t)$

(c) $-4t^2\cos(2t)$

(d) $-2s^2\sin(2t)$

(e) $-6s^2\sin(2t)$

6. Consider the unit vector $u = \dfrac{1}{\sqrt{2}}(i + j)$ and the two functions

$$f(x,y) = \begin{cases} \dfrac{4xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 2, & (x,y) = (0,0) \end{cases} \qquad g(x,y) = \begin{cases} \dfrac{4xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

Which of the following is FALSE?

(a) $g_x(0,0) = 0$

(b) g is not differentiable at $(0,0)$

(c) $D_u g(0,0)$ does not exist

(d) $D_u f(0,0) = 0$

(e) f is differentiable at $(0,0)$

7. Which one of the following statements is FALSE?

(a) The graph of $16x^2 + y^2 + 16z^2 = 4$ is a cylinder

(b) The graph of $16x^2 - y^2 + 16z^2 = 0$ is an elliptic cone

(c) The graph of $16x^2 - y^2 + 16z^2 = 4$ is a hyperboloid of one sheet

(d) The graph of $16x^2 - y + 16z^2 = 4$ is an elliptic paraboloid

(e) The graph of $16x^2 - y^2 - 16z^2 = 4$ is a hyperboloid of two sheets

8. The domain of the function $f(x,y) = \sqrt{16 - x^2 - 4y^2}$ is the set of all points

(a) lying on or inside the circle $x^2 + y^2 = 16$

(b) lying on or inside the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{4} = 1$

(c) lying on or inside the ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{2} = 1$

(d) lying outside the ellipse $\dfrac{x^2}{4} + \dfrac{y^2}{16} = 1$

(e) lying on or outside the circle $x^2 + y^2 = 16$

9. If $z = \ln(4x^2 + y^2)$, then $x\dfrac{\partial z}{\partial x} + y\dfrac{\partial z}{\partial y} =$

(a) z^2

(b) 0

(c) 2

(d) $-z$

(e) z

10. An equation of the tangent plane to the surface

$$z = y\, e^{2xy}$$

at the point $(0, 2, 2)$ is

(a) $y - 8z = 0$

(b) $x + y - 8z = 0$

(c) $8x + y - z = 0$

(d) $8x + y + z = 0$

(e) $32x + y - z = 0$

11. If z is defined implicitly as a differentiable function of x and y by the equation

$$x \ln y + y^2 z + z^2 = 6,$$

then $\dfrac{\partial z}{\partial y}$ at the point $(6, 1, 2)$ is equal to

(a) $\dfrac{9}{5}$

(b) $-\dfrac{9}{4}$

(c) $-\dfrac{5}{3}$

(d) -2

(e) $\dfrac{27}{19}$

12. Let $z = x^2 + y^2$. If (x, y) changes from $(1, 1)$ to $(1.3, 0.9)$, then using differentials, the change in z is approximately equal to

(a) 0.4

(b) -0.1

(c) 0.05

(d) 0.2

(e) 0.02

13. The maximum value of the directional derivative of $f(x, y) = y^2 - x\sqrt{y}$ at the point $(4, 4)$ is equal to

(a) 7

(b) $\sqrt{53}$

(c) 3

(d) 4

(e) $\sqrt{33}$

14. If $ax + by + cz = 8$ is an equation for the plane passing through the points $P(2, 1, 0)$ and $Q(3, 0, 4)$ and is perpendicular to the yz-plane, then $a + b + c =$

(a) 10

(b) 6

(c) 3

(d) 4

(e) 8

15. $\lim_{(x,y)\to(0,0)} \dfrac{4x^3}{2x^2+2y^2} =$

(a) $\dfrac{3}{2}$

(b) 2

(c) 1

(d) DNE

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$t = 1$, $s = 2$: point $(3, -2, -3)$, $abc = 18$.
2	**(c)**	$f_{xyzz} = 4y = 8$ and $f_{yzxz} = 4y = 8$; sum 16.
3	**(d)**	$f(0, 0, 3) = -9$, so $4x^2 + 4y^2 - z^2 = -9$.
4	**(a)**	Normals $\langle 3, -2, 1\rangle$ and $\langle 6, -4, -2\rangle$ are not parallel; the other statements hold.
5	**(b)**	$w_t = 4x(-s\sin t) - 2y(s\cos t) = -6s^2 \sin t \cos t = -3s^2 \sin 2t$.
6	**(e)**	The limit at the origin does not exist, so f is not continuous, hence not differentiable.
7	**(a)**	$16x^2 + y^2 + 16z^2 = 4$ is an ellipsoid, not a cylinder.
8	**(b)**	$16 - x^2 - 4y^2 \geq 0 \iff \dfrac{x^2}{16} + \dfrac{y^2}{4} \leq 1$.
9	**(c)**	$xz_x + yz_y = \dfrac{8x^2 + 2y^2}{4x^2 + y^2} = 2$.
10	**(c)**	$f_x(0, 2) = 8$, $f_y(0, 2) = 1$: $z - 2 = 8x + (y - 2)$, i.e. $8x + y - z = 0$.
11	**(d)**	$F_y = 10$, $F_z = 5$, so $z_y = -F_y/F_z = -2$.
12	**(a)**	$dz = 2(0.3) + 2(-0.1) = 0.4$.
13	**(b)**	$\nabla f(4, 4) = \langle -2, 7\rangle$, maximum $\sqrt{53}$.
14	**(a)**	$a = 0$; P gives $b = 8$, Q gives $c = 2$; sum 10.
15	**(e)**	Polar: $2r\cos^3\theta \to 0$.